Talk 10: Partial Trace & Tensor Tricks
Quantum Information Theory

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Overview

- Quick recap of notation
- Useful Identities
- Practice problems

Notation recap

Let H_A and H_B be finite-dimensional complex Hilbert spaces.

- Vectors are kets, $|x\rangle \in H_A$.
- Tensor product: $H_A \otimes H_B$
- Operators: $\mathcal{L}(H_A)$, $\mathcal{L}(H_A \otimes H_B)$
- An operator $A \in \mathcal{L}(H)$ maps:

$$\begin{aligned} A : H_A &\to H_A \\ |x\rangle &\mapsto A|x\rangle \end{aligned}$$

- Partial trace over B:

$$\mathrm{tr}_B : \mathcal{L}(H_A \otimes H_B) \to \mathcal{L}(H_A)$$

Identity 1: Vector push out

VECTOR PUSHOUT

For kets:

$$\begin{aligned}(|a\rangle \otimes |b\rangle) &= (I_A \otimes |b\rangle)|a\rangle \qquad \mathbb{C} \otimes \mathbb{C} \cong \mathbb{C} \to H_A \otimes H_B \\ &= (|a\rangle \otimes I_B)|b\rangle\end{aligned}$$

For bras:

$$\begin{aligned}(\langle a| \otimes \langle b|) &= \langle a|(I_A \otimes \langle b|) \qquad H_A \otimes H_B \to \mathbb{C} \otimes \mathbb{C} \cong \mathbb{C} \\ &= \langle b|(\langle a| \otimes I_B)\end{aligned}$$

Identity 1: Vector push out, cont'd

WHY? $(|A\rangle \otimes |B\rangle) = (I_A \otimes |B\rangle)|A\rangle : \mathbb{C} \otimes \mathbb{C} \cong \mathbb{C} \to H_A \otimes H_B$

Background recall:

- A ket and a bra as operators acts as:

$$\begin{aligned} |a\rangle : \mathbb{C} &\to H_A & \langle a| : H_A &\to \mathbb{C} \\ c &\mapsto c|a\rangle & |x\rangle &\mapsto \langle a|x\rangle \end{aligned}$$

- These are all equivalent:

$$|a\rangle = 1 \cdot |a\rangle = 1_{\mathbb{C}} \otimes |a\rangle = |a\rangle \otimes 1_{\mathbb{C}} = |a\rangle(1)$$

- the LHS $|a\rangle \otimes |b\rangle$ maps:

$$\mathbb{C} \otimes \mathbb{C} \to H_A \otimes H_B$$

- So, $|a\rangle \otimes |b\rangle$, acts on two scalars by,

$$(|a\rangle \otimes |b\rangle)(c_1 \otimes c_1) = c_1|a\rangle \otimes c_2|b\rangle = c_1c_2(|a\rangle \otimes |b\rangle)$$

- For the RHS we have ($\Rightarrow$ recall $(A \otimes B)(C \otimes D) = AC \otimes BD$):

$$\begin{aligned} (I_A \otimes |b\rangle)|a\rangle &= (I_A \otimes |b\rangle)(|a\rangle \otimes 1_{\mathbb{C}}) \\ &= I_a|a\rangle \otimes |b\rangle 1_{\mathbb{C}} \\ &= |a\rangle \otimes |b\rangle \end{aligned}$$

- Which equals the LHS. $\square$

Identity 2: Operator Pushout

OPERATOR PUSHOUT

For kets:

$$(A \otimes I)(I \otimes |b\rangle) = (I \otimes |b\rangle)A: \qquad H_A \to H_A \otimes H_B$$
$$(I \otimes B)(|a\rangle \otimes I) = (|a\rangle \otimes I)B: \qquad H_B \to H_A \otimes H_B$$

For bras:

$$(I \otimes \langle b|)(A \otimes I) = A(I \otimes \langle b|): \qquad H_A \otimes H_B \to H_A$$
$$(\langle a| \otimes I)(I \otimes B) = B(\langle a| \otimes I): \qquad H_A \otimes H_B \to H_B$$

Identity 2: Operator push out, cont'd

WHY? $(\mathbf{I} \otimes \langle \mathbf{B}|)(\mathbf{A} \otimes \mathbf{I}) = \mathbf{A}(\mathbf{I} \otimes \langle \mathbf{B}|) : H_A \otimes H_B \to H_A$

- Operate on $|x\rangle \otimes |y\rangle$ on both the left and right hand sides.
- **LHS:**

$$\begin{aligned}(I \otimes \langle b|)(A \otimes I) \circ (|x\rangle \otimes |y\rangle) &= (I \otimes \langle b|)(A|x\rangle \otimes |y\rangle) \\ &= IA|x\rangle \otimes \langle x|y\rangle \\ &= \langle x|y\rangle \cdot A|x\rangle\end{aligned}$$

- **RHS:**

$$\begin{aligned}A(I \otimes \langle b|) \circ (|x\rangle \otimes |y\rangle) &= A(I|x\rangle \otimes \langle b|y\rangle) \\ &= A(\langle b|y\rangle \cdot |x\rangle) \\ &= \langle x|y\rangle \cdot A|x\rangle\end{aligned}$$

- LHS = RHS. $\square$
- The other operator push out identities are similar.

Partial Trace in Practice

Goals

- Understand **what** the partial trace is conceptually.
- Learn **how to compute** tr_A and tr_B **by hand**.
- Practice with **small explicit examples**:
  - Generic 4×4 matrix (two qubits).
  - Bell state density matrix.
  - Product pure state.
  - Unequal dimensions $d_A \neq d_B$.

Setup:

- Global space: $\mathcal{H}_A \otimes \mathcal{H}_B$.
- Dimensions: $\dim \mathcal{H}_A = d_A$, $\dim \mathcal{H}_B = d_B$.
- Fix an orthonormal product basis

$$\{\,|i\rangle_A\}_{i=0}^{d_A-1}, \quad \{\,|j\rangle_B\}_{j=0}^{d_B-1}, \quad \{|i\rangle_A \otimes |j\rangle_B\}.$$

What the Partial Trace Does

Conceptual Picture

Given an operator X on $\mathcal{H}_A \otimes \mathcal{H}_B$:

- $\mathbf{tr}_A \boldsymbol{X}$:
  The operator on B obtained by "discarding" or "tracing out" system A.
- $\mathbf{tr}_B \boldsymbol{X}$:
  The operator on A obtained by "discarding" or "tracing out" system B.

In Terms of Expectation Values

- For any observable M on A and density matrix ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$,

$$\mathrm{tr}\left(M\mathrm{tr}_B(\rho_{AB})\right) = \mathrm{tr}\left((M \otimes \mathrm{I}_B)\rho_{AB}\right).$$

- For any observable N on B and density matrix ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$,

$$\mathrm{tr}\left(N\mathrm{tr}_A(\rho_{AB})\right) = \mathrm{tr}\left((\mathrm{I}_A \otimes N)\rho_{AB}\right).$$

! PRESERVATION OF STATISTICS

This is the formal statement that the partial trace “preserves all measurement statistics” on the subsystem you keep.

Index Formulae for $\mathbf{tr}_A$ and $\mathbf{tr}_B$

Coordinate Notation

- Use basis $\{|i\rangle_A \otimes |a\rangle_B\}$ with indices:
- i, j for system A (range $0, \ldots, d_A - 1$),
- a, b for system B (range $0, \ldots, d_B - 1$).
- Write matrix elements of X as

$$X_{ia,\,jb} = \langle i, a | X | j, b \rangle.$$

Index Formulae for Matrix Elements

$$(\mathrm{tr}_A X)_{ab} = \sum_{i=0}^{d_A - 1} X_{ia,\,ib}$$

$$(\mathrm{tr}_B X)_{ij} = \sum_{a=0}^{d_B - 1} X_{ia,\,ja}$$

EASY CALCULATION

These are often the easiest formulae to check calculations or prove identities.

Block–Matrix Recipes

A-major Ordering and Block Structure

- Assume we order the basis of $A \otimes B$ by $\boldsymbol{A}$ **first, then** $\boldsymbol{B}$ (A-major):

$$|0_A 0_B\rangle,\ |0_A 1_B\rangle,\ \ldots,\ |1_A 0_B\rangle,\ |1_A 1_B\rangle,\ \ldots$$

- Then X is a $d_A d_B \times d_A d_B$ matrix that we view as a $d_A \times d_A$ grid of $d_B \times d_B$ blocks:

$$X = \begin{bmatrix} X_{00} & X_{01} & \cdots & X_{0,d_A-1} \\ X_{10} & X_{11} & \cdots & X_{1,d_A-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{d_A-1,0} & X_{d_A-1,1} & \cdots & X_{d_A-1,d_A-1} \end{bmatrix}, \quad X_{ij} \in \mathbb{C}^{d_B \times d_B}.$$

Block Recipes

$$\mathrm{tr}_A X = \sum_{i=0}^{d_A - 1} X_{ii}$$

$$\mathrm{tr}_B X = \left[\ \mathrm{tr}(X_{ij})\ \right]_{i,j=0}^{d_A - 1}$$

FAST HAND CALCULATION

These block rules are usually the **fastest** way to do partial traces by hand for small systems.

Example 1: Two Qubits, Generic 4×4 Matrix

Matrix and Block Decomposition

- Let $d_A = d_B = 2$. In the A-major ordering $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, take

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}.$$

- Block this as 2×2 blocks of size 2×2:

$$X = \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{array}\right] = \begin{bmatrix} X_{00} & X_{01} \\ X_{10} & X_{11} \end{bmatrix}.$$

Example 1 (cont.): Compute $\mathbf{tr}_A$ and $\mathbf{tr}_B$

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} = \begin{bmatrix} X_{00} & X_{01} \\ X_{10} & X_{11} \end{bmatrix}.$$

Trace Out A

- Sum the diagonal blocks:

$$\mathrm{tr}_A X = X_{00} + X_{11} = \begin{bmatrix} 1+11 & 2+12 \\ 5+15 & 6+16 \end{bmatrix} = \begin{bmatrix} 12 & 14 \\ 20 & 22 \end{bmatrix}.$$

- This is a 2×2 operator on B.

Trace Out B

- Take traces of each block:

$$\mathrm{tr}_B X = \begin{bmatrix} \mathrm{tr} X_{00} & \mathrm{tr} X_{01} \\ \mathrm{tr} X_{10} & \mathrm{tr} X_{11} \end{bmatrix} = \begin{bmatrix} (1+6) & (3+8) \\ (9+14) & (11+16) \end{bmatrix} = \begin{bmatrix} 7 & 11 \\ 23 & 27 \end{bmatrix}.$$

- This is a 2×2 operator on A.

Example 2: Bell State $|\Phi^+\rangle$

Definition and Density Matrix

- Consider the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big), \quad \rho_{AB} = |\Phi^+\rangle\langle\Phi^+|.$$

- In the A-major basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$,

$$\rho_{AB} = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

Example 2 (cont.): Block Form and Partial Traces

Block Decomposition

$$\rho_{AB} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} X_{00} & X_{01} \\ X_{10} & X_{11} \end{bmatrix}.$$

Compute the Partial Traces

- **Trace out A:**

$$\mathrm{tr}_A \rho_{AB} = X_{00} + X_{11} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \frac{1}{2} I_2.$$

- **Trace out B:**

$$\mathrm{tr}_B \rho_{AB} = \begin{bmatrix} \mathrm{tr} X_{00} & \mathrm{tr} X_{01} \\ \mathrm{tr} X_{10} & \mathrm{tr} X_{11} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \frac{1}{2} I_2.$$

> **NOTE**
>
> Each subsystem is **maximally mixed**, as expected for a maximally entangled state.

Example 3: Product Pure State

Setup

- Let

$$|a\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |b\rangle = \gamma|0\rangle + \delta|1\rangle.$$

- The joint pure state is

$$|\psi\rangle = |a\rangle \otimes |b\rangle, \quad \rho_{AB} = |\psi\rangle\langle\psi| = (|a\rangle\langle a|) \otimes (|b\rangle\langle b|).$$

- Using linearity and $\mathrm{tr}(|b\rangle\langle b|) = 1$,

$$\mathrm{tr}_B \rho_{AB} = |a\rangle\langle a| \cdot \mathrm{tr}(|b\rangle\langle b|) = |a\rangle\langle a|.$$

- Similarly,

$$\mathrm{tr}_A \rho_{AB} = \mathrm{tr}(|a\rangle\langle a|)\, |b\rangle\langle b| = |b\rangle\langle b|.$$

- So for a product pure state, the reduced states are just the original pure states of each subsystem.
- *(You can also expand ρ_{AB} into a 4×4 matrix and apply the block rules to see this explicitly.)*

Example 4: Unequal Dimensions $d_A = 3, d_B = 2$

Block Structure

Now $\dim\mathcal{H}_A = 3$, $\dim\mathcal{H}_B = 2$, so X is a 6×6 matrix.

Write it as a 3×3 grid of 2×2 blocks:

$$X = \begin{bmatrix} X_{00} & X_{01} & X_{02} \\ X_{10} & X_{11} & X_{12} \\ X_{20} & X_{21} & X_{22} \end{bmatrix}, \quad X_{ij} \in \mathbb{C}^{2\times 2}.$$

Partial Traces in This Case

- **Trace out A:**

$$\mathrm{tr}_A X = X_{00} + X_{11} + X_{22},$$

  which is still a 2×2 operator on B.

- **Trace out B:**

$$\mathrm{tr}_B X = \left[\, \mathrm{tr}(X_{ij}) \,\right]_{i,j=0}^{2},$$

  which is a 3×3 operator on A.

The structure is exactly the same; only the block grid size changes.

Sanity Checks and Common Pitfalls

Sanity Checks

- **Dimension check**:
  - $\mathrm{tr}_A X$ is $d_B \times d_B$.
  - $\mathrm{tr}_B X$ is $d_A \times d_A$.
- **Trace preservation**:

$$\mathrm{tr}(\mathrm{tr}_A X) = \mathrm{tr}(\mathrm{tr}_B X) = \mathrm{tr}(X)$$

- **Linearity**:

$$\mathrm{tr}_A(\alpha X + \beta Y) = \alpha\, \mathrm{tr}_A X + \beta\, \mathrm{tr}_A Y,$$

and similarly for tr_B.

Common Pitfalls

- **Not using a product basis**:
  The simple "sum diagonal blocks" rule **assumes** a product basis ordered A-major (or B-major, if you swap roles).
- **Mixing up which system you trace out**:
  In A-major ordering, summing the diagonal $d_B \times d_B$ blocks gives tr_A, not tr_B.
- **Forgetting which indices belong to which system**:
  Always keep track of i, j for A and a, b for B when using

$$(\mathrm{tr}_A X)_{ab} = \sum_i X_{ia,\ ib}, \qquad (\mathrm{tr}_B X)_{ij} = \sum_a X_{ia,\ ja}.$$

Summary Slide

Key Recipes to Remember

Matrix Elements

$$(\mathrm{tr}_A X)_{ab} = \sum_{i=0}^{d_A-1} X_{ia,\ ib}$$

$$(\mathrm{tr}_B X)_{ij} = \sum_{a=0}^{d_B-1} X_{ia,\ ja}$$

Block-wise Calculation

$$\mathrm{tr}_A X = \sum_{i=0}^{d_A-1} X_{ii}$$

$$\mathrm{tr}_B X = \left[\ \mathrm{tr}(X_{ij})\ \right]_{i,j}$$

> **TIP**
>
> With these, you can reliably compute partial traces by hand for small systems and check more abstract derivations.